Forecasting Recent College Graduate Unemployment

An analytical deep dive into unemployment trends among recent college graduates using a custom SARIMA model.

View GitHub Repository.

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Exploratory Data Analysis: Setting the Stage

Our journey began with comprehensive data collection from diverse sources to understand the landscape of recent college graduate unemployment.

Federal Reserve Economic Data (FRED)

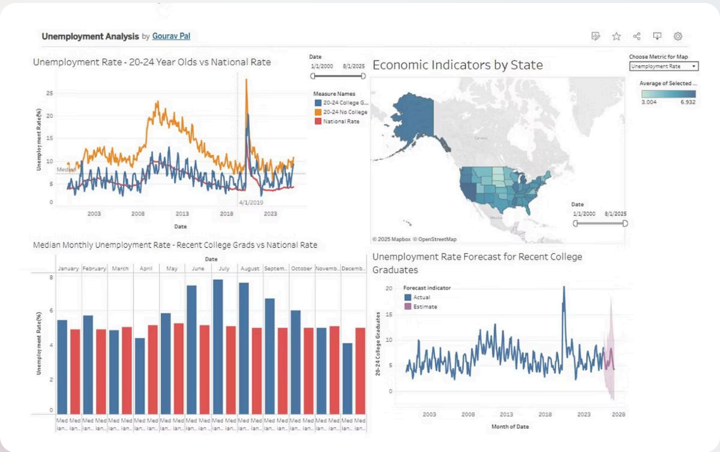
Utilized Python to pull data on unemployment rates for college graduates (Bachelor's Degree, 20-24 years), national unemployment rates, and non-college graduate rates for the same age group.

Series Link: Unemployment Rate College Graduates

State-Level Economic Indicators

Web scraping techniques were employed to gather monthly data (2000-2025) on Unemployment, Labor Force Participation Rate, and Minimum Wage from all 50 states, totaling 145 different series.

Note: Alabama, Louisiana, Mississippi, South Carolina, and Tennessee lack state-specific minimum wage data.



Visualizing the Data: An Interactive Dashboard

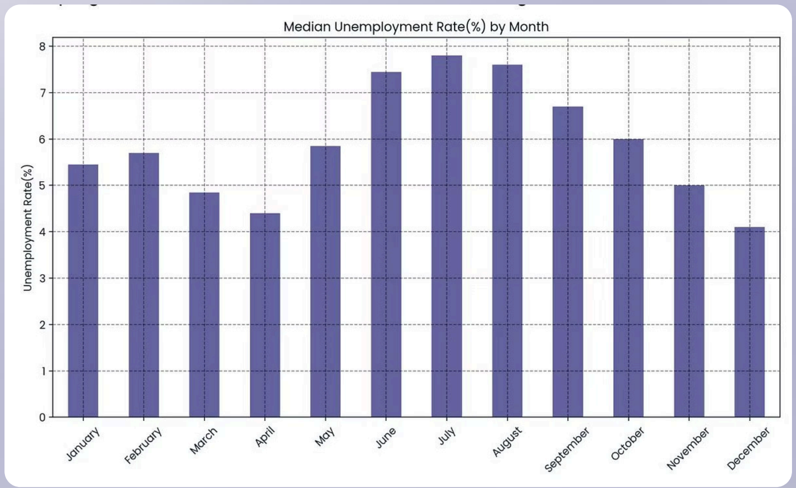
All collected data was integrated into an interactive Tableau dashboard, revealing crucial patterns and shifts in unemployment rates.

> This dashboard served as the foundation for identifying key insights and anomalies in the labor market for recent graduates.

View Interactive Dashboard

Key Insights: Shifting Dynamics & Seasonal Patterns

Analysis of the interactive dashboard highlighted two significant trends:



National vs. Graduate Unemployment

Historically, the National Unemployment Rate mirrored that of recent college graduates. However, since 2019, the recent college graduate unemployment rate has consistently remained **above** the national average for over three years.

Pronounced Seasonal Peaks

A strong seasonal pattern exists for recent graduates, with unemployment peaking in summer due to new entrants in the job market, then decreasing in winter and spring as graduates secure employment.

Correlation Analysis: Unpacking Economic Factors

Examining the relationships between unemployment, labor force participation, and minimum wage across states provided unexpected results.

Unemployment & Minimum Wage

Correlation: -0.22

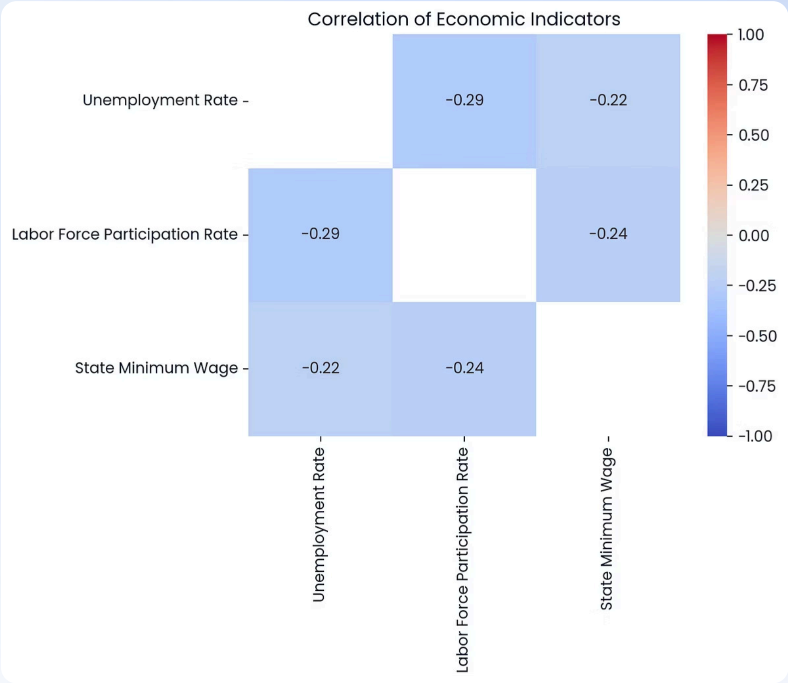
Unemployment & Labor Force Participation

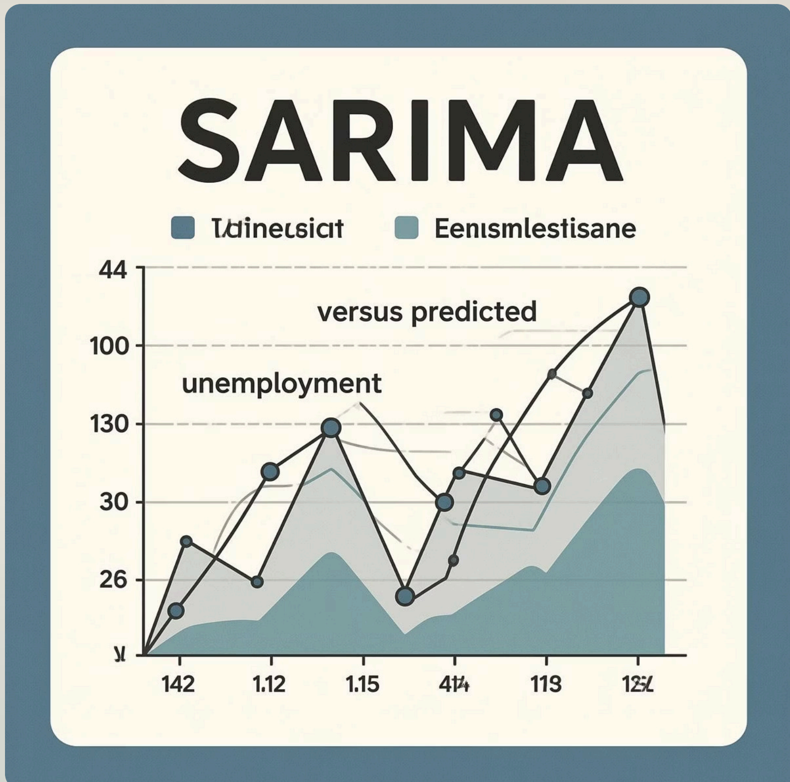
Correlation: -0.28

Labor Force Participation & Minimum Wage

Correlation: -0.24

These values suggest no statistically significant relationship among these variables within the collected data (2000-2025).





SARIMA Model: Predictive Power

To forecast future unemployment rates, a custom Seasonal AutoRegressive Integrated Moving Average (SARIMA) model was developed in Python.

1. **Model Customization**

   Leveraged Python for greater customizability compared to standard in-built features in platforms like Tableau.

2. **Training & Validation**

   The model was trained on data from 2000-2021 and validated against a set from 2021-August 1, 2025.

3. **Rolling Forecast Origin**

   Evaluated using a rolling forecast origin method, making one-step-ahead predictions and continuously updating the training set with actual values for robust validation.

SARIMA Model Performance & Forecast

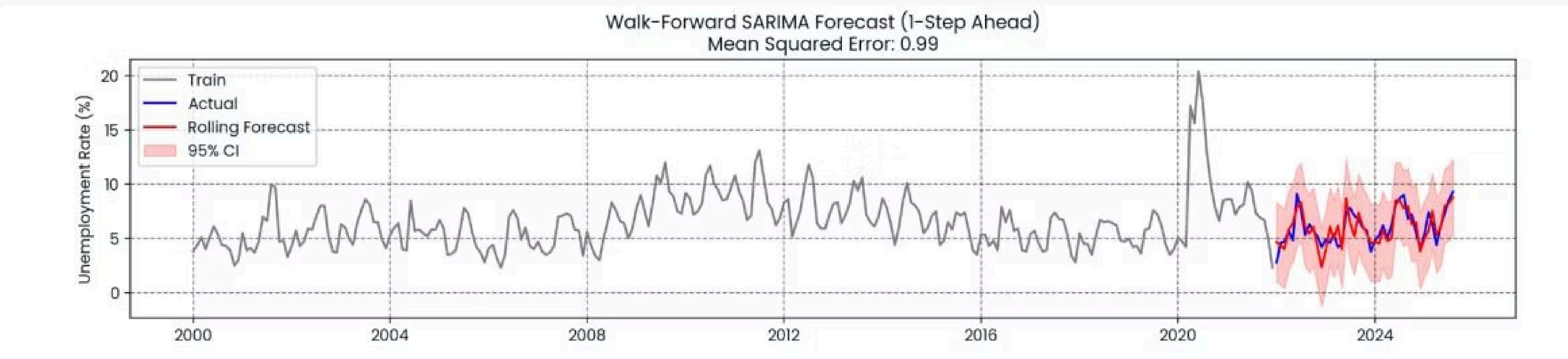
The model demonstrated strong performance on validation and provided a clear outlook for future unemployment trends.

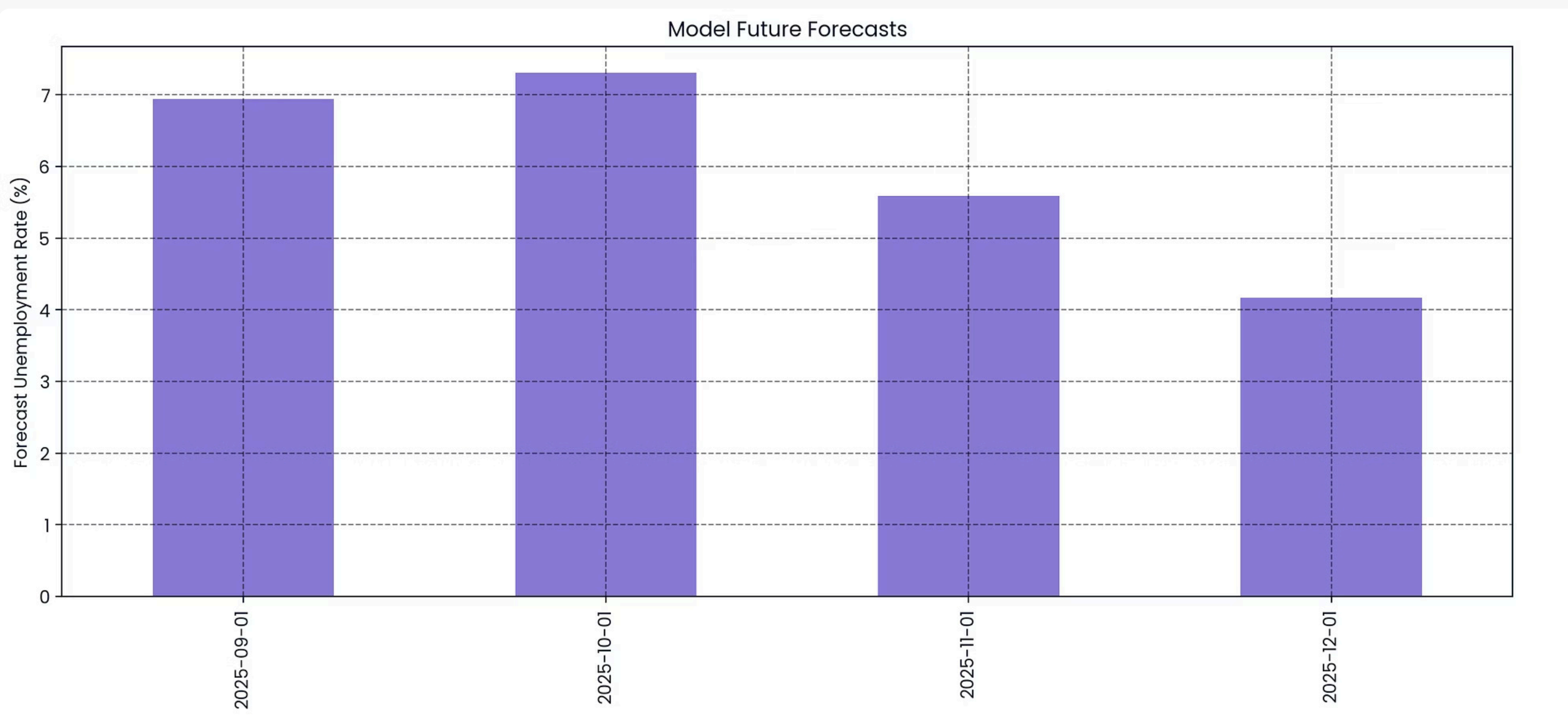
Validation Success

Achieved a Mean Squared Error (MSE) score of 0.99, indicating a very good fit for the scale of the data.

Confidence Intervals

The model's confidence intervals consistently enveloped the actual unemployment rates, reinforcing its reliability.





SARIMA Model Summary

```
                               SARIMAX Results
==========================================================================================
Dep. Variable:                                    y   No. Observations:                  308
Model:             SARIMAX(1, 0, 1)x(0, 1, [], 12)   Log Likelihood                -586.439
Date:                              Sat, 25 Oct 2025   AIC                           1178.877
Time:                                      15:55:08   BIC                           1189.948
Sample:                                  01-01-2000   HQIC                          1183.310
                                       - 08-01-2025
Covariance Type:                                opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1          0.7839      0.040     19.712      0.000       0.706       0.862
ma.L1         -0.1843      0.082     -2.255      0.024      -0.344      -0.024
sigma2         3.0718      0.080     38.452      0.000       2.915       3.228
===================================================================================
Ljung-Box (L1) (Q):                   0.12   Jarque-Bera (JB):              4052.13
Prob(Q):                              0.72   Prob(JB):                         0.00
Heteroskedasticity (H):               2.35   Skew:                             0.69
Prob(H) (two-sided):                  0.00   Kurtosis:                        21.07
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
```

Future Forecast Summary (2025)

Date	predicted_mean	lower y	upper y
2025-09-01	6.94	3.51	10.38
2025-10-01	7.31	3.31	11.32
2025-11-01	5.59	1.27	9.9
2025-12-01	4.17	-0.33	8.67

Update: September 2025 Unemployment Rate

The September 2025 unemployment rate for college graduates came in higher than anticipated, indicating a divergence from typical seasonal patterns.

Actual vs. Forecast

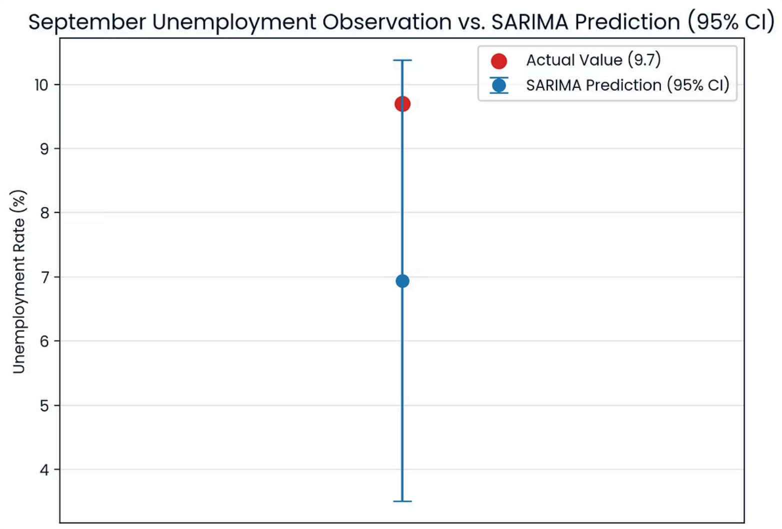
The actual rate for September was 9.7%, exceeding the model's expectation of a drop to approximately 7%.

Historically, September typically sees a rate of 6.7% for recent college graduates.

Model Resilience

Despite the deviation, the SARIMA model's confidence intervals still successfully encompassed the actual rate, demonstrating its robust predictive range.

This unexpected increase suggests evolving economic conditions are impacting the entry-level job market.



Longest Increase Streak: A Concerning Trend

Further analysis revealed a significant and rare occurrence in the unemployment rate for recent college graduates.

1

Historic Precedent

The current five-month consecutive increase in unemployment rates for recent college graduates has only occurred once before, in March 2007.

2

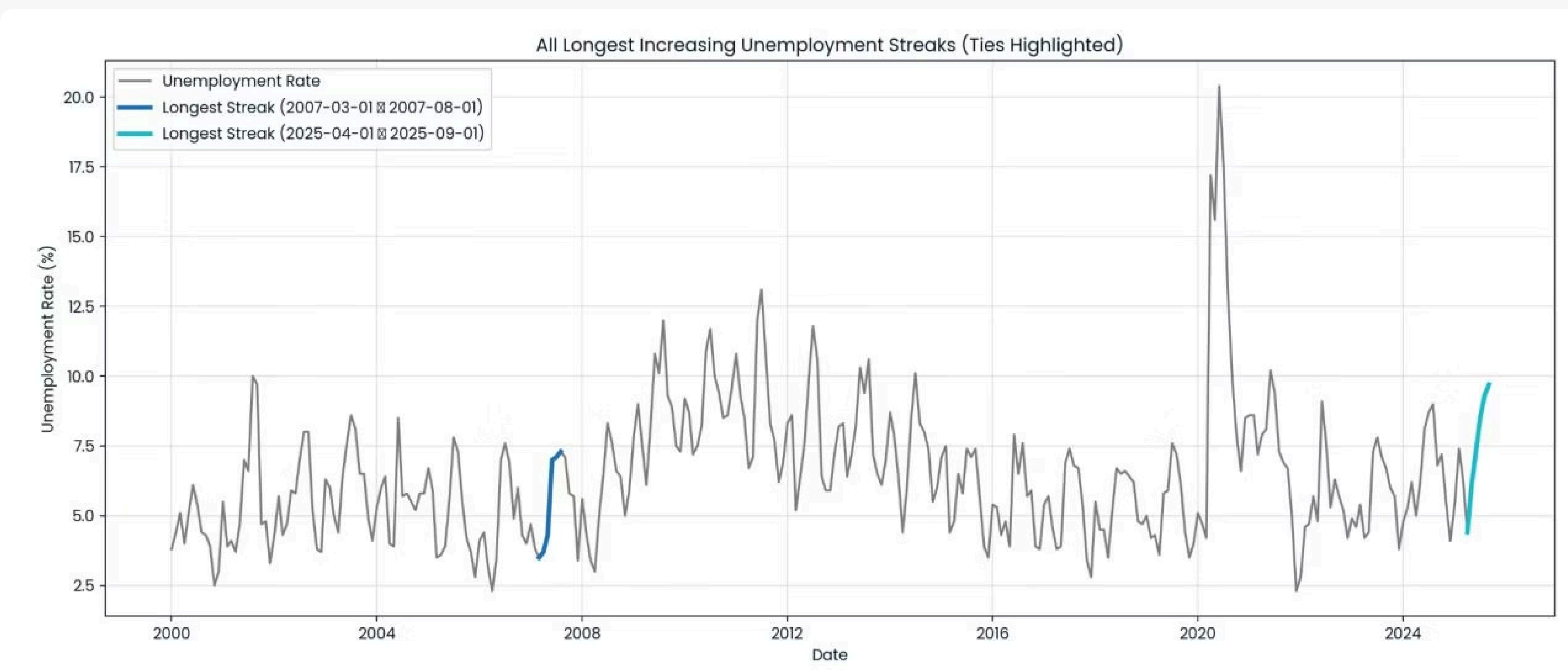
Slowing Job Market

This extended streak signals a potential slowdown in the job market, particularly for recent college graduates and entry-level positions.

Python analysis identified two notable streaks:

```
Longest increase streak length: 6 months
Number of tied streaks: 2
---- Streak #1 ----
Start: 2007-03-01
End: 2007-08-01

---- Streak #2 ----
Start: 2025-04-01
End: 2025-09-01
```



Potential Factors Contributing to Rising Unemployment

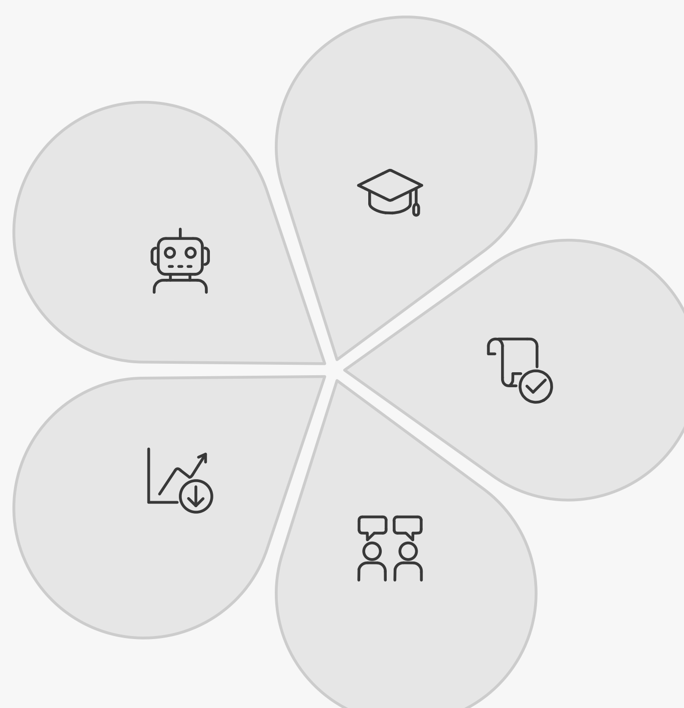
Several macroeconomic and societal factors could be influencing the current increase in recent college graduate unemployment.

Graduate Over-supply

An increasing number of college graduates may be outstripping the demand for entry-level work. As much as 52% of college graduates are underemployed upon initial labor market entry. They are likely to stay that way—even 10 years after graduation, 45% of college graduates are underemployed, according to the Talent Disrupted report - Federal Reserve - August, 2025

AI Advancements

Artificial Intelligence may be replacing entry-level roles traditionally filled by recent graduates. Job listings for entry-level corporate roles have declined 15%, while applications per job have surged 30%, showing AI's squeeze on new grads. - CBS News - July, 2025



Degree Value Erosion

Grade inflation and less competitive college environments might diminish the perceived value of a degree. College GPAs rose more than 16% from 1990-2020, with "A" now the most common grade awarded - US Department of Education - September, 2025

Economic Uncertainty

Broader economic concerns, tariffs, and restrictive regulations can slow hiring across sectors.

Political Polarization

Political instability and divided governance can lead to economic uncertainty, impacting job creation.

Understanding these intertwined factors is crucial for addressing the challenges faced by recent college graduates in the evolving labor market.